

# Immigration, Gender, and Citizenship: Intersecting Inequalities in the U.S. Labor Market

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## Abstract

This study examines wage disparities by gender and citizenship status using pooled cross-sectional data from the 2012–2019 American Community Survey. Wage differentials are estimated using a modified Mincer earnings equation with human capital and race controls, along with year, industry, birthplace, and geographic fixed effects. Two-fold pooled Blinder–Oaxaca decompositions are then used to quantify the sources of group wage gaps. Conditional on human capital, race, and fixed effects, women earn approximately 30% less than men, and the decomposition attributes 92% of the raw gender gap to differences in returns rather than differences in characteristics. The citizen–non-citizen gap exhibits the opposite pattern: 86% is explained by observable human capital, primarily English proficiency and education. At the intersection of the two dimensions, non-citizen women face a statistically significant additional wage penalty of approximately 4.7% beyond the independent associations of gender and citizenship, and within the non-citizen workforce, 94% of the gender gap remains unexplained. These findings document the compounded disadvantage faced by non-citizen women and underscore the importance of intersectional labor market policies.

**Keywords:** Wage gap · Citizenship · Gender wage differential · Immigrant · Intersectionality

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All remaining errors are my own.

## I Introduction

It is often said that when the U.S. catches a cold, Mexico gets pneumonia. A reflection of the economic interdependence between Mexico and the U.S. (Malkin, 2008). Yet, is immigration truly detrimental to the U.S. economy? If so, why has the U.S. implemented policies such as the Bracero Program, which allowed Mexicans to work in the U.S. for up to six months? (US House of Representatives: History, Art & Archives, nd; Massey et al., 2002). The truth is that there is nothing more permanent than a temporary work permit.

The foreign-born population in the United States reached a record high of 53.3 million in January 2025, representing 15.8% of the total population (Pew Research Center, 2025). Of these, Mexican immigrants comprise the largest group at 21% of all U.S. immigrants (Pew Research Center, 2025). Among the total immigrant population, approximately 11 million are unauthorized (Baker and Warren, 2024). Despite immigrants' role in the U.S. economy, particularly in labor-intensive industries, they often face significant barriers and are perceived by policymakers as what I will hereafter term an “undesired necessity” (National Immigration Forum, 2017).

The debate over immigration's economic impact remains contentious among economists. The Mariel boatlift of 1980, which brought 120,000 low-skilled Cuban migrants to Miami, exemplifies this disagreement. While National Academies of Sciences, Engineering, and Medicine (2017) found little evidence that this influx harmed native workers' wages or employment, Borjas (2017) argued it significantly depressed low-skilled native wages. Card and Peri (2016) later challenged these findings, concluding there was “essentially no effect” on Miami's labor market.

Rather than engaging in this broader debate, this paper focuses on a critical yet understudied aspect: the intersection of citizenship status and gender in wage determination. Specifically, it examines how these factors shape wage disparities between native-born citizens, naturalized citizens, and non-citizens across the U.S. labor market. With this research, I seek to answer the following question: *How do citizenship status and sex shape wage disparities in the U.S. labor market?* This paper analyzes pooled cross-sectional individual-level data from the 2012–2019 American Community Survey (ACS). Geographic diversity is ensured through a 5% single-stage cluster sample of Public Use Microdata Areas (PUMAs). I estimate wage differentials using a modified Mincer earnings equation that includes human capital and race controls, along with year, industry, birthplace,

and geographic fixed effects, and I quantify the sources of the resulting wage gaps using two-fold pooled Blinder–Oaxaca decompositions. The remainder of the paper proceeds as follows: Section II reviews the literature, Section III presents the theory, Section IV describes the data, Section V outlines the empirical strategy, Section VI reports the results, and Section VII concludes.

## II Literature Review

While there is dispute over immigration’s broader impact on the U.S. economy, there is consensus on one observation: a wage gap between immigrant and native workers (Borjas, 1994; Coulombe et al., 2014; Smith and Fernandez, 2017). Scholars continue to debate the relative weight of the following three wage determinants:

### 1. Human Capital

Immigrants’ wage disadvantages are often attributed to disparities in human capital. Borjas (1994) argues that U.S.-born workers inherently benefit from English proficiency, which allows immigrants to escape “ethnic enclaves” and access broader opportunities (Garg, 2004). Education also plays a critical role: immigrants complete 1.3 fewer years of formal education than natives and are far more likely to lack a high school diploma (Camarota and Zeigler, 2016; Card, 2005). Even when immigrants possess comparable qualifications, foreign credentials are systematically devalued. Chiswick (1978) found that immigrants receive lower returns to foreign education and experience, a pattern Blau and Kahn (2015) confirmed, showing U.S. employers discount “non-domestic skill signals” for newly legalized immigrants.

However, human capital alone cannot fully explain persistent gaps. Coulombe et al. (2014) find that once human capital quality is accounted for, the unexplained portion of the immigrant–native wage gap decreases (by 62% for men and is nearly eliminated for women). Donahue (2021) reinforces this critique, finding that wage gaps persist even among immigrants with 25+ years of U.S. residency, suggesting structural barriers like unfair labor policies or discrimination. Coulombe et al.’s study is used as a conceptual guide rather than as an empirical precedent for the U.S., as their study is based in Canada.

## 2. Occupational Segregation

Immigrants face systemic barriers that funnel them into low-wage industries. Borjas (2018) attributes this to “imperfect substitutability,” arguing that immigrants cluster in occupations natives avoid. Similarly, Smith and Fernandez (2017) found that immigrants disproportionately work in low-wage sectors, while Garg (2004) emphasizes how limited English proficiency traps them in industries like construction, where wages lag by 12–18%.

Occupational segregation is not voluntary. Elliott and Lindley (2008), Blau and Kahn (2015), and Ottaviano and Peri (2012) argue that limited education restricts immigrants to unskilled jobs, but this ignores gender dynamics. Vernez (1999) notes that female immigrants face “additional cultural and social barriers,” such as childcare responsibilities and spousal income effects (Schoeni, 1998). A husband’s higher wages may reduce a woman’s labor force participation (income effect) or increase her household productivity (substitution effect).

## 3. Discrimination

Discrimination is a persistent yet underaddressed factor. Oreopoulos (2011) attributes wage disparities to “various intangible factors,” including employer bias. Hersch (2011) quantifies colorism, showing that immigrants with darker skin tones earn 16–23% less than lighter-skinned peers, indicating that wage gaps persist beyond human capital and reflect racialized judgments similar to those tied to non-citizen status. These penalties intersect with gender. Vernez (1999) documents compounded barriers for immigrant women, and Garg (2004) shows Latina workers concentrated in the sub-\$35,000 bracket compared to Asian counterparts. This highlights how racist and sexist biases persistently shape economic outcomes and research.

Citizenship status further exacerbates these inequities. The National Academies of Sciences, Engineering, and Medicine (2017) stress that unauthorized workers face heightened vulnerability, yet no study compares naturalized and non-citizen women’s exposure to discrimination. Massey et al. (2002) and National Immigration Forum (2017) emphasize industries like agriculture, where non-citizens dominate, but they fail to analyze how citizenship-gender intersections shape outcomes.

Taken together, existing research leaves critical gaps that this paper aims to address directly. First, while scholars like Borjas (1994) and Blau and Kahn (2015) emphasize credential devaluation,



no work examines whether naturalization improves returns to foreign education. Second, though Vernez (1999) and Garg (2004) identify gendered barriers for female immigrants, they overlook how citizenship status, particularly the distinction between naturalized and non-citizen women, shapes occupational mobility out of “ethnic enclaves” (Garg, 2004). Third, discrimination frameworks (Hersch, 2011; Oreopoulos, 2011) treat immigrants as uniform groups, failing to capture how non-citizen women face compounded penalties distinct from naturalized peers.

This study addresses these gaps by disaggregating citizenship status to examine whether legal status moderates the returns to human capital. It also models intersectional wage penalties through gender–citizenship interactions estimated alongside industry fixed effects, building on Donahue’s (2021) finding that “structural problems” perpetuate disparities even among equally qualified workers. A Blinder–Oaxaca decomposition isolates the role of discrimination, extending Oreopoulos’s (2011) focus on “unobserved factors” to quantify bias in non-citizen women’s wage outcomes. By analyzing the full breadth of industries within an economy-wide framework, this paper provides new evidence on how citizenship and gender jointly shape wage inequality, addressing the National Academies of Sciences, Engineering, and Medicine’s (2017) call for intersectional analyses of immigrant labor market outcomes.

### III Economic Theory

Human capital theory (Becker, 1964; Mincer, 1974) argues that wages reflect investments in education, skills, and experience. However, immigrants face credential devaluation, where foreign education and work experience are systematically discounted by employers (Chiswick, 1978; Blau and Kahn, 2015).

Taste-based discrimination (Becker, 1957) extends this by framing employer prejudice as a willingness to sacrifice profits to avoid hiring certain groups. Employers act as if hiring non-citizens, women, or both imposes a nonpecuniary cost, quantified by a discrimination coefficient  $d$ . The perceived cost of employing a member of a disfavored group thus becomes  $w(1 + d)$ , exceeding their market wage unless the worker accepts a wage discount.<sup>1</sup> Credential devaluation exacerbates this: foreign degrees, especially from non-Western institutions, are treated as noisy or unreliable

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<sup>1</sup>In Becker’s model,  $w$  is the worker’s market wage and  $d$  is the discrimination coefficient. If  $d > 0$ , the employer perceives hiring costs as  $w(1 + d)$ , making the worker less competitive unless they accept a wage discount.

signals of productivity (Banerjee and Lee, 2015; Li, 2001). Donahue (2021), for example, finds that immigrants with foreign credentials earn 18% less than natives with comparable U.S. qualifications, even after controlling for occupation.

Statistical discrimination complicates this dynamic. Employers use group averages, based on citizenship status and/or gender, to proxy unobserved productivity under imperfect information (Aigner and Cain, 1977). Non-citizens may be stereotyped as unreliable due to visa instability and women as less committed due to caregiving roles, compounding penalties for non-citizen women.<sup>2</sup> While taste-based discrimination reflects explicit prejudice, statistical discrimination rationalizes bias through perceived risk; however, both ultimately devalue human capital.

Immigration policies create institutional barriers for non-citizens, limiting labor protections and access to high-wage sectors (Massey et al., 2002). Unauthorized workers lack minimum wage protections in agriculture, while temporary visa holders remain dependent on employers. Naturalization mitigates these barriers by granting legal protections. However, non-citizens and especially women remain concentrated in industries with weak labor enforcement, including domestic work, seasonal agriculture, and services. While unionized sectors may offer narrower wage gaps through collective bargaining, non-citizen women are often excluded from membership (National Immigration Forum, 2017). Thus, Institutional Theory suggests that naturalized citizens likely experience narrower wage gaps compared to natives, while non-citizens face persistent penalties, especially in informal sectors.

Intersectionality theory argues that overlapping identities create multiplicative, not additive, disadvantages (Browne and Misra, 2003). Non-citizen women face “double jeopardy:” credential devaluation (human capital) and stereotypes about caregiving roles (statistical discrimination). For instance, Latina immigrants are often channeled into low-wage care work, where employers assume they will tolerate subminimum wages (Garg, 2004).<sup>3</sup> This intersectional framework suggests non-citizen women face wage penalties exceeding the combined effects of gender and citizenship status

<sup>2</sup>For example, if employers statistically associate unauthorized status with job turnover, they may avoid investing in training for non-citizen women, relegating them to informal roles (Massey et al., 2002).

<sup>3</sup>This illustrates Becker’s framework,  $w(1 + d)$ . A higher  $d$  inflates the perceived cost of hiring, leading employers to offer lower wages or exclude Latinas entirely.

alone. Hence, by considering these four frameworks, I model wages as a function of:

$$Wage = f(\text{Human Capital, Discrimination, Institutional Barriers, Intersectional Identity}) \quad (1)$$

These four frameworks generate directly testable predictions. If human capital theory alone governed wages, group gaps would vanish once education, English proficiency, and experience are held constant. The discrimination and intersectionality frameworks instead predict that a penalty for non-citizen women survives such controls and that the unexplained share of the decomposition is largest for that group. My hypotheses are therefore the following:

$H_0$  : Once human capital differences are accounted for, there is no wage gap between non-citizen women and their counterparts across gender, citizenship status, and their interaction.

$H_A$  : Wage gaps persist after controlling for human capital, with non-citizen women facing compounded penalties beyond the independent effects of gender and citizenship alone.

Based on human capital theory and taste-based discrimination, I expect non-citizen women to exhibit systematically lower returns to education than their naturalized or native-born peers, and based on statistical discrimination, I expect the unexplained component of the Blinder–Oaxaca decomposition to be largest for non-citizen women, reflecting employer reliance on group stereotypes rather than individual productivity.

## IV Data Description

This study analyzes individual-level American Community Survey (ACS) data from the years 2012 up to 2019, a period selected to ensure consistency in Public Use Microdata Area (PUMA) boundaries.<sup>4</sup> The unit of observation is the individual worker: each record is one employed, wage-earning adult aged 16–66 drawn from the pooled 2012–2019 cross-sections. ACS interviews a new sample of households each year and does not follow individuals over time. The raw extract comprises 25.3 million observations. I then draw a 5% simple random sample of PUMA–state geographies and retain every individual living in a sampled geography, yielding a single-stage cluster sample. After removing invalid income and age records and restricting to employed wage earners, the final sample

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<sup>4</sup>See Ruggles et al. (2025) for data source.

includes 526,832 individuals across 118 unique PUMA–state geographies. Because sampling occurs at the geography level, all inference clusters standard errors on these 118 geographies, and the models employ PUMA–state fixed effects to account for localized economic conditions. Appendix Figure 10 maps the sampled geographies: coverage spans all four census regions and includes both metropolitan and rural areas, which supports exploiting spatial variation through PUMA–state fixed effects and geography-clustered inference.

The dataset captures variables for analyzing labor market inequalities: citizenship status (categorized as native citizen, naturalized citizen, or non-citizen), sex, industry, and wage income (adjusted to 2015 dollars).<sup>5</sup> Educational attainment is grouped into three categories: high school or less (up to and including grade 12), some college (one to three years of college, no bachelor’s degree), and bachelor’s degree or higher. This study defines immigrants as individuals born in a foreign country. Naturalized citizens are categorized separately to examine how the naturalization process may influence wage disparities. Table 4 shows that 85.17% of the sample are U.S.-born citizens, while 14.83% are foreign-born, including 8.25% naturalized citizens and 6.59% non-citizens. Gender distribution is nearly equal (51.49% male, 48.51% female), though wage disparities are stark: men earn an average of \$62,537 compared to women’s \$41,441. These averages reflect selection into employment, as the sample captures only employed, wage-earning workers.

Table 5 presents initial evidence of wage differences across groups.<sup>6</sup> Notably, naturalized citizens earn higher mean annual wages (\$58,556) than native-born citizens (\$52,517), suggesting positive selection into naturalization. Second, non-citizens have the lowest mean wages (\$41,713) and exhibit a pronounced gender gap: non-citizen women earn just 66% of their male counterparts’ wages (\$31,864 vs. \$48,422).<sup>7</sup>

Non-citizen workers are notably concentrated in agriculture, construction, and key service industries. Table 7 shows that service sectors alone absorb a substantial share: Professional Services accounts for 13.77%, Arts for 13.26%, and Other Services for 5.51% of non-citizen employment. Agriculture & Forestry employs 2.55%, while Construction represents 10.08%. Together, these five

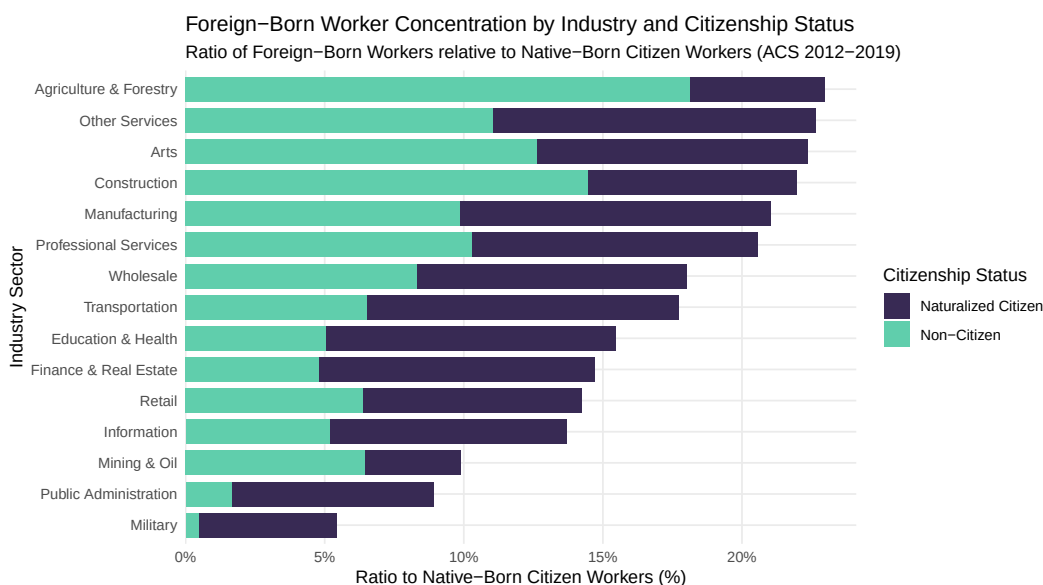
<sup>5</sup>All wages and household income are first converted to 1999 dollars via IPUMS’s CPI99, then adjusted to 2015 dollars by multiplying by 1/0.703, where 0.703 is the CPI99 index value for 2015.

<sup>6</sup>Mean wages are sensitive to high earners; see Table 6 for median wages. Medians are notably lower across all groups, most starkly for non-citizens (\$26,000 median vs. \$41,713 mean), suggesting right-skewed distributions.

<sup>7</sup>ACS does not distinguish between legal and undocumented immigration statuses. Thus, this category serves as a catch-all that likely includes unauthorized workers, which may partially drive these lower wage estimates.

sectors comprise 45.17% of all non-citizen workers. Figure 1 further illustrates that both naturalized and non-citizen workers are disproportionately represented in labor-intensive, lower-wage industries.<sup>8</sup>

Fig. 1: Ratio of Foreign-Born to Citizen Workers by Industry



*Note:* Each bar reports the number of naturalized-citizen and non-citizen workers per native-born citizen worker.

The dataset also reflects demographic diversity of the foreign-born population. As shown in Figure 2, most immigrants originate from Mexico, Southeast Asia, and Europe. Mexico was assigned its own category due to its large representation in the dataset.

Figure 3 displays the distribution of income levels within educational groups, disaggregated by gender and citizenship. Even among workers holding a bachelor's degree or higher, non-citizen women remain disproportionately represented in the low-income bracket. By contrast, native-born men show the most favorable income distribution, especially at higher educational levels.

<sup>8</sup>Ratios were calculated by dividing the number of foreign-born workers in each industry by the number of U.S.-born citizen workers in the same industry.

Fig. 2: Demographic Diversity by Birthplace

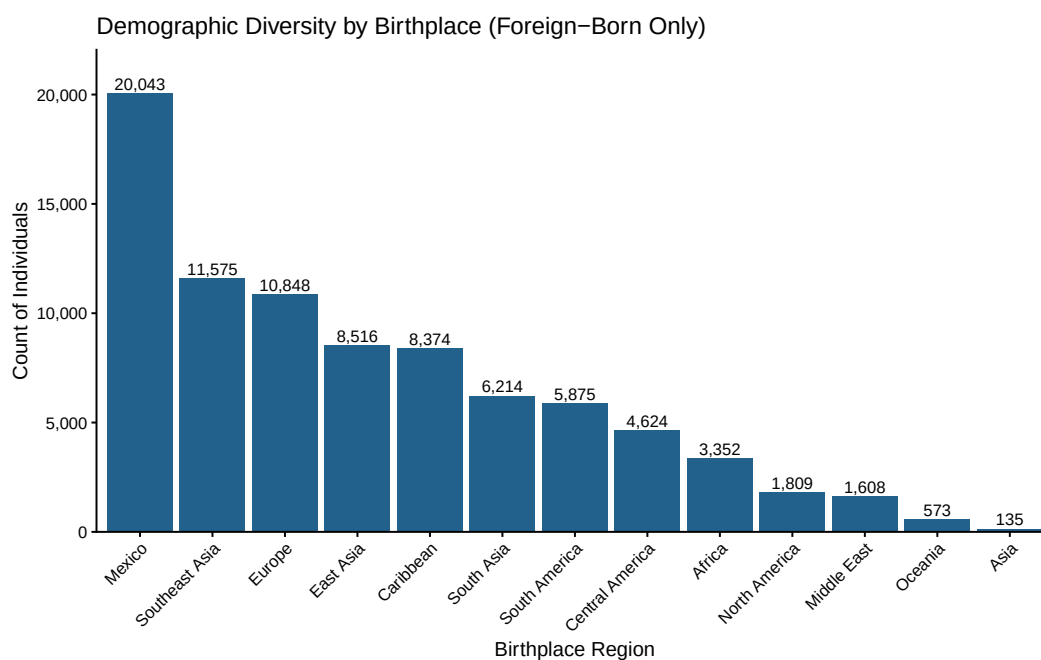
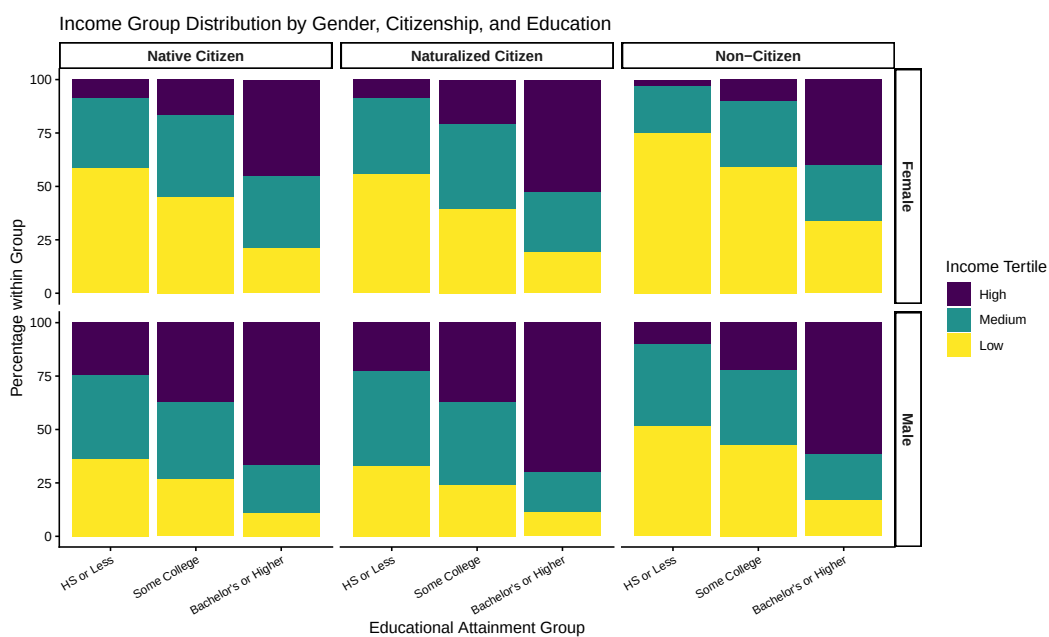


Fig. 3: Income Group Distribution by Gender, Citizenship, and Education



## V Empirical Strategy

I utilize two econometric approaches to explore the wage differentials between citizenship status and sex, focusing on isolating the role of discrimination:

### A Mincer Earnings Function

I begin with a modified Mincer wage equation to capture the returns to human capital and other observable characteristics. Let  $i$  index individuals,  $s$  the PUMA–state geography of residence, and  $t$  the survey year. The population regression model is

$$\begin{aligned} \ln(wage_{i,s,t}) = & \beta_0 + \beta_1 Nat_{i,s,t} + \beta_2 NonCit_{i,s,t} + \beta_3 Female_{i,s,t} \\ & + \beta_4 (Nat \times Female)_{i,s,t} + \beta_5 (NonCit \times Female)_{i,s,t} \\ & + \mathbf{Z}'_{i,s,t} \boldsymbol{\gamma} + \alpha_s + \tau_t + \theta_{j(i)} + \psi_{b(i)} + \varepsilon_{i,s,t} \end{aligned} \quad (2)$$

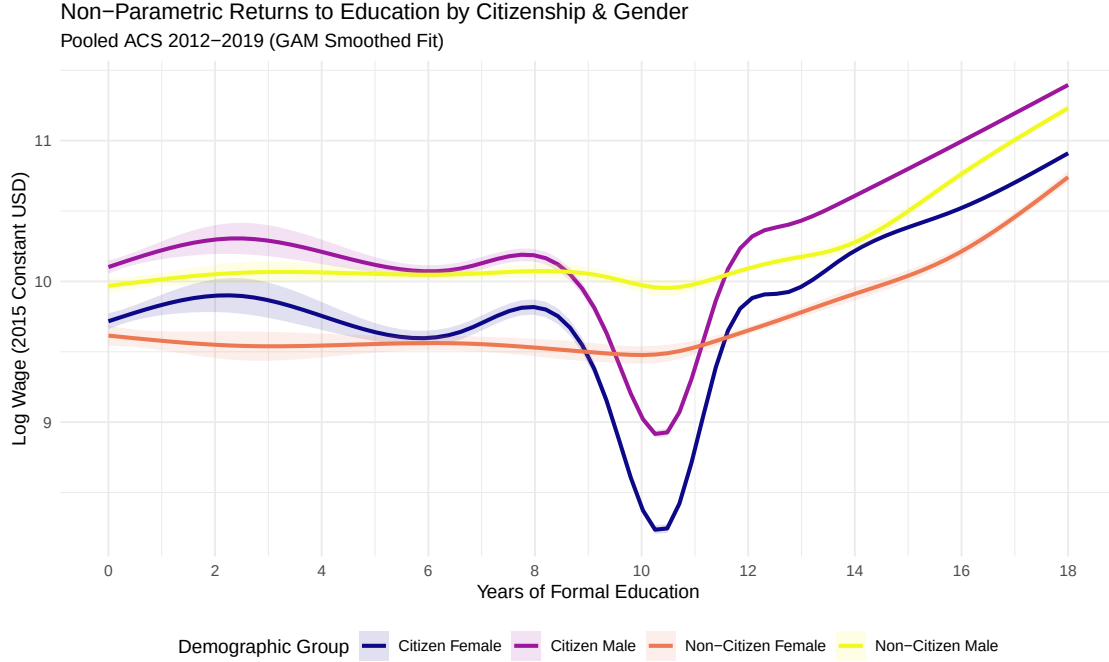
where  $\ln(wage_{i,s,t})$  is the log annual wage (in constant 2015 dollars) of individual  $i$  in geography  $s$  and year  $t$ .<sup>9</sup>  $Nat_{i,s,t}$  and  $NonCit_{i,s,t}$  are indicators for naturalized citizens and non-citizens (native-born citizens are the omitted reference group),  $Female_{i,s,t}$  indicates women (men are the reference), and the two interaction terms allow the association between citizenship and wages to differ by sex;  $\beta_5$  is the parameter of central interest, measuring the *additional* wage gap faced by non-citizen women beyond the separate gender and citizenship terms. Because Equation (2) is the population model, its parameters carry no hats; hats ( $\hat{\beta}$ ) denote the corresponding OLS estimates reported in the tables.

The vector of individual-level controls,  $\mathbf{Z}'_{i,s,t}$ , includes age polynomials and human capital controls. Within this vector, education enters nonlinearly through indicators for three harmonized attainment groups: high school or less, some college, and bachelor’s degree or higher. A single linear years-of-schooling term would force each additional year to carry the same proportional wage gain, but returns to education are convex and concentrated at credential thresholds—the “sheep-skin” effects associated with degree completion (Hungerford and Solon, 1987; Lemieux, 2006). The categorical specification allows the data to price the high school-to-some-college and some-college-

<sup>9</sup>The log transformation reduces the skewness of the wage distribution from 4.22 to  $-1.09$ ; see Appendix Figure 9.

to-degree steps freely. Figure 4 supports this choice: non-parametric (GAM) fits of log wages on years of education steepen visibly beyond twelve years and around degree completion for every citizenship–gender group.

Fig. 4: Non-Parametric Returns to Education by Citizenship and Gender



*Note:* The convex profiles motivate the categorical education specification in Equation (2).

The remaining controls within  $\mathbf{Z}_{i,s,t}$  include a quadratic age term centered at its sample mean;<sup>10</sup> this quadratic captures the concave experience profile of the standard Mincer equation. The vector  $\mathbf{Z}_{i,s,t}$  also contains additional demographic controls: English proficiency (fluent, intermediate, limited), number of resident children (none, one, two, three–four, five or more), and race from the IPUMS classification (White as the reference; Black, American Indian/Alaska Native, Asian/Pacific Islander, and Hispanic/Latino).

The terms  $\alpha_s$ ,  $\tau_t$ ,  $\theta_{j(i)}$ , and  $\psi_{b(i)}$  are fixed effects for PUMA–state geography, survey year, industry, and birthplace region, respectively, absorbing unobserved geographic, temporal, sectoral, and origin-country heterogeneity.  $\varepsilon_{i,s,t}$  collects all unobserved wage determinants. Standard errors are clustered on the 118 PUMA–state geographies in every specification.

<sup>10</sup>Centering reduces the mechanical collinearity between age and its square (all variance inflation factors are below 2), so the linear age coefficient measures the wage–age gradient at the average age rather than at age zero.



A design choice that merits explanation is the inclusion of race indicators *alongside* birthplace fixed effects. These two are complements, not substitutes. Birthplace fixed effects absorb origin-country factors—the quality and transferability of source-country schooling, origin-specific networks, and selection into migration—that are common to everyone born in a given region. Race, by contrast, captures racial identification and its treatment in the U.S. labor market: racial stratification and colorism operate *within* and *across* origin groups (Hersch, 2011), so two workers from the same birthplace region (or two native-born workers) may face different labor market treatment by race. Omitting race would also load racial wage differentials onto the citizenship terms to the extent that race and citizenship covary. For example, 13.1% of the sample identifies as Hispanic/Latino, a group disproportionately represented among non-citizens. Because racial categories may reflect both unequal treatment and unobserved sorting, I interpret the race coefficients as conditional associations consistent with racial stratification, not as causal effects of race.

A deliberate feature of Equation (2) is the *exclusion* of household income. Household income is a classic “bad control” (Angrist and Pischke, 2009): an individual’s own wage is a component of their household’s income, so the causal chain runs

$$\text{Citizenship} \longrightarrow \text{Individual Wage} \longrightarrow \text{Household Income}.$$

Conditioning on household income would therefore partially control for the dependent variable itself, inducing post-treatment bias and an endogenous selection of comparisons (workers with equal household income but different wages must differ in offsetting household characteristics). For this reason household income appears in no regression or decomposition in this paper; at most it could be entertained in a supplementary robustness specification.

The specification is estimated identically in Stata (`reghdfe`) and R (`fixest`); the two implementations produce numerically identical coefficient estimates, which serves as a reproducibility check.

## B Blinder–Oaxaca Decomposition

I then apply a Blinder–Oaxaca decomposition (Blinder, 1973; Oaxaca, 1973) to four dyadic contrasts: (1) the overall gender gap, (2) the citizenship gap, (3) the gap between non-citizen and

citizen women, and (4) the gap between non-citizen women and non-citizen men. I use the *two-fold* decomposition with pooled reference coefficients (Neumark, 1988; Jann, 2008). Writing  $\Delta = E[\ln(wage) | A] - E[\ln(wage) | B]$  for the mean log wage gap between groups  $A$  and  $B$ , the decomposition is

$$\Delta = \underbrace{[E(X_A) - E(X_B)]' \beta^*}_{\text{explained}} + \underbrace{[E(X_A)'(\beta_A - \beta^*) + E(X_B)'(\beta^* - \beta_B)]}_{\text{unexplained}} \quad (3)$$

where  $X_g$  is the vector of characteristics and  $\beta_g$  the coefficient vector for group  $g \in \{A, B\}$ , and  $\beta^*$  is the reference coefficient vector estimated from the pooled sample of both groups, including a group-membership indicator to avoid contaminating the pooled slopes with the group gap itself (Jann, 2008). The *explained* (endowments) component values the groups' differences in observable characteristics—education, English proficiency, age, children, year, and industry composition—at the pooled returns  $\beta^*$ : it is the part of the gap attributable to differences in productive characteristics. The *unexplained* component captures differences in the *returns* to those characteristics, i.e., different slopes and intercepts for the two groups relative to the pooled benchmark. In the labor economics tradition this component proxies for labor market discrimination, though it also absorbs any group differences in unobserved productivity determinants, so I interpret it as an upper bound on discrimination rather than a point measure.

The decomposition uses the same controls as Equation (2) with three exceptions. First, the 118 geographic fixed effects are excluded for dimensionality reasons: the decomposition estimates separate models within each group, and the full dummy set cannot be reliably estimated within the smaller groups (e.g., 14,059 non-citizen women). Second, birthplace fixed effects are excluded because birthplace is group-defining in the citizenship contrasts—essentially all non-citizens are foreign-born—so treating it as a “characteristic” would mechanically absorb the group difference being decomposed. Third, the race indicators are excluded from the decompositions: the explained component values differences in characteristics as legitimate productivity differentials, so treating racial composition as an “endowment” would reclassify race-linked pay differences—which may themselves reflect differential treatment—as explained, understating the disparity attributable to unequal returns. Excluding race keeps race-linked differentials inside the unexplained component, consistent with its interpretation as an upper bound on discrimination. Year and industry indicators

are retained in every decomposition, and standard errors are again clustered on PUMA–state geographies.

I acknowledge that the non-citizen category includes both unauthorized workers and temporary visa holders like international students and H-1B/H-2A workers,<sup>11</sup> who differ in legal rights and labor market access. Despite the robustness of my models, unobserved factors, such as ability, visa type nuances, social networks, health status, or union membership, may still bias the estimates. The race measure is likewise coarse: it is a mutually exclusive, self-identified classification that cannot capture skin-tone gradients within categories, which Hersch (2011) shows matter for immigrant wages. Because the data are pooled cross-sections, the estimates should be read as conditional associations rather than causal effects of citizenship or gender. Equally, the Oaxaca “unexplained” component, while serving as a proxy for discrimination, also captures residual heterogeneity. Thus, my results should be interpreted not as definitive causal effects, but as approximations.

Building on Oreopoulos (2011) and Donahue (2021), this approach uses a Mincer regression to model baseline wage determinants and a Blinder–Oaxaca decomposition to quantify how much of the remaining gap, particularly for non-citizen women, reflects structural barriers like discrimination.

## VI Results

### A Regression Results

Table 1 presents the regression results examining the association between gender, citizenship, and log wages across three model specifications.<sup>12</sup> Each model progressively refines the estimates by controlling for human capital characteristics and structural factors. All three models cluster standard errors on the 118 PUMA–state geographies.

Model 1 captures the raw associations without human capital or structural controls. Naturalized citizens earn 16.0% more than natives,<sup>13</sup> while non-citizens earn 21.9% less, reflecting unadjusted

<sup>11</sup>H-1B visas are granted to high-skilled specialty workers, typically to international students after receiving their college degree. Generally, H-1B lasts a year, but for STEM fields it can last up to 3 years. H-2A visas are issued to temporary agricultural workers for seasonal farming jobs. Since the abolition of the Bracero Program, temporary agricultural workers have been admitted with H-2A visas.

<sup>12</sup>Models are reported unweighted. As a robustness check, weighted models using ACS person weights (`perwt`) are reported in Appendix Tables 13 and 14 and yield similar results.

<sup>13</sup>Coefficients on indicator variables in a log wage equation are converted to percentage differentials as  $(e^{\hat{\beta}} - 1) \times 100$ ;

structural and human capital disparities. Women earn 33.2% less than men, underscoring systemic gender inequality. Non-citizen women earn approximately 3.9% less still, beyond the separate gender and citizenship terms. Although parsimonious, this model already provides initial evidence of intersectional disadvantage for non-citizen women.

Table 1: Effect of Gender and Citizenship on Log Wages

| Variables                           | Model 1              | Model 2              | Model 3              |
|-------------------------------------|----------------------|----------------------|----------------------|
| Naturalized Citizen (NC)            | 0.148***<br>(0.028)  | 0.042***<br>(0.013)  | 0.045**<br>(0.019)   |
| Non-Citizen (NotC)                  | -0.247***<br>(0.030) | -0.018<br>(0.017)    | -0.020<br>(0.020)    |
| Female                              | -0.403***<br>(0.012) | -0.437***<br>(0.012) | -0.359***<br>(0.009) |
| Naturalized Citizen $\times$ Female | 0.019<br>(0.021)     | 0.043**<br>(0.021)   | 0.023<br>(0.018)     |
| Non-Citizen $\times$ Female         | -0.040**<br>(0.018)  | -0.042***<br>(0.016) | -0.048***<br>(0.014) |
| Constant                            | 10.583***<br>(0.024) | 10.591***<br>(0.015) | 10.577***<br>(0.007) |
| Observations                        | 526,832              | 526,832              | 526,832              |
| Controls                            |                      | X                    | X                    |
| Fixed Effects                       |                      |                      | X                    |
| $R^2$                               | 0.038                | 0.339                | 0.390                |

Standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Model 1: Raw associations between citizenship, gender, and log wages; no controls. Model 2: Adds human capital and demographic controls. Model 3: Adds fixed effects for year, industry, birthplace, and geography. Full coefficient estimates for all models appear in Appendix Table 12.

Model 2 shows that the citizenship main effects vanish once human capital and race are held constant: the naturalized premium falls from 16.0% to 4.3%, and the non-citizen penalty shrinks from 21.9% to a statistically insignificant 1.8%, confirming that the raw citizenship gaps stem principally from differences in education, English proficiency, age structure, and racial composition. The gender penalty, by contrast, *widens* to 35.4%. This is revealing: women in the sample hold slightly *better* observable characteristics than men (a pattern the decomposition below confirms), so comparing men and women with equal credentials enlarges rather than shrinks the adjusted gap. The additional penalty for non-citizen women persists at approximately 4.2% ( $p < 0.01$ ).

for small coefficients the log-point and percentage readings nearly coincide.

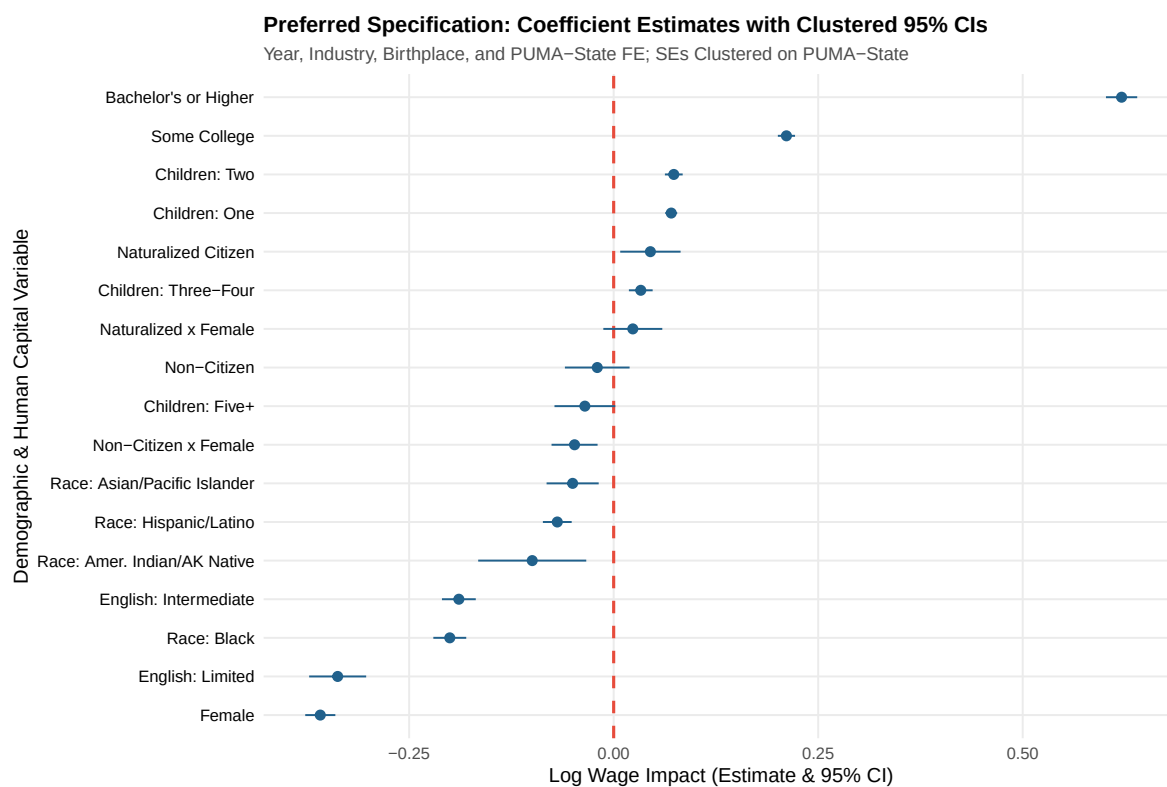
Model 3, the preferred specification, adds fixed effects for year, industry, birthplace, and PUMA—state geography, absorbing unobserved heterogeneity across time, sectors, origin countries, and local labor markets. The naturalization premium is 4.6% ( $p < 0.05$ ), aligning with Borjas (1994) and Donahue (2021), who argue that legal status mitigates structural barriers but does not fully erase disparities. The non-citizen main effect is statistically indistinguishable from zero, which contrasts with Coulombe et al. (2014) and Smith and Fernandez (2017) and suggests that, conditional on human capital, race, industry, and location, non-citizen *men* do not face a measurable penalty; sectoral sorting noted by Massey et al. (2002) does the work instead. The gender penalty remains 30.1% ( $p < 0.01$ ), underscoring the systemic devaluation of women’s labor consistent with Hersch (2011) and Garg (2004). Critically, the “double disadvantage” for non-citizen women—an additional wage penalty of approximately 4.7% ( $p < 0.01$ )—survives the full fixed-effects battery, reflecting the multiplicative intersectional disadvantage theorized by Browne and Misra (2003) and the compounded barriers documented by Vernez (1999).

The race coefficients (Appendix Table 12) document racial stratification conditional on citizenship, education, industry, location, and birthplace: Black workers earn approximately 18.2% less than observationally similar White workers, American Indian and Alaska Native workers 9.5% less, Hispanic/Latino workers 6.7% less, and Asian/Pacific Islander workers 4.9% less (all  $p < 0.01$ ). Because birthplace fixed effects are held constant, these differentials compare workers from the same origin region and are consistent with the racial hierarchy and colorism documented by Hersch (2011). Although these estimates should not be interpreted as causal effects of race and may partly reflect unobserved sorting across occupations and firms, they suggest that racial disparities operate alongside, rather than instead of, differences associated with citizenship status and gender.

Figure 5 visualizes the full preferred specification. The education coefficients rise steeply and convexly—some college raises wages by approximately 23.5% relative to high school or less, while a bachelor’s degree or higher raises them by 86.1%—consistent with the sheepskin effects motivating the categorical education specification. The English proficiency gradient is similarly steep: intermediate speakers earn 17.2% less and limited speakers 28.6% less than fluent speakers, echoing Garg’s (2004) emphasis on language as a gateway out of ethnic enclaves.

Selection effects likely shape several of these estimates. Naturalized citizens must pass English and civics tests and maintain steady employment, so those who naturalize are a positively selected,

Fig. 5: Preferred Specification: Coefficient Estimates with Clustered 95% Confidence Intervals



*Note:* Point estimates and 95% confidence intervals from Model 3. Age terms are omitted from the display.  
 Reference categories: native citizen, male, high school or less, fluent English, no children, White.

higher-ability group. This selection inflates the raw naturalization premium in Model 1 (16.0%) but largely vanishes once education, age, race, and fixed effects are controlled (4.6%). Conversely, the non-citizen category blends unauthorized workers in low-wage, high-risk jobs with H-1B or H-2A visa holders in higher-paid sectors, so its estimates mix actual returns with sorting across visa classes.

These findings align with the statistical discrimination framework outlined earlier. Employers may use citizenship status as a proxy for unobserved productivity, particularly for women. The persistence of the negative interaction term for non-citizen women across all three models reveals how employer bias can amplify their wage disadvantage.

## B Blinder–Oaxaca Decomposition

Table 2 reports the two-fold pooled decomposition of each of the four contrasts into an explained component (differences in characteristics, valued at pooled returns) and an unexplained component (differences in returns to those characteristics). Detailed decompositions with grouped covariate contributions appear in Appendix B.

Table 2: Blinder–Oaxaca Decompositions: Two-Fold, Pooled Reference

| Contrast                     | Total Gap            | Explained            | Unexplained          | % Unexplained |
|------------------------------|----------------------|----------------------|----------------------|---------------|
| Male vs. Female              | 0.397***<br>(0.010)  | 0.031***<br>(0.005)  | 0.366***<br>(0.008)  | 92.1          |
| Citizen vs. Non-Citizen      | 0.243***<br>(0.023)  | 0.208***<br>(0.021)  | 0.035**<br>(0.014)   | 14.3          |
| NC Female vs. Citizen Female | −0.302***<br>(0.024) | −0.219***<br>(0.022) | −0.084***<br>(0.015) | 27.7          |
| NC Female vs. NC Male        | −0.443***<br>(0.017) | −0.025**<br>(0.010)  | −0.417***<br>(0.015) | 94.3          |

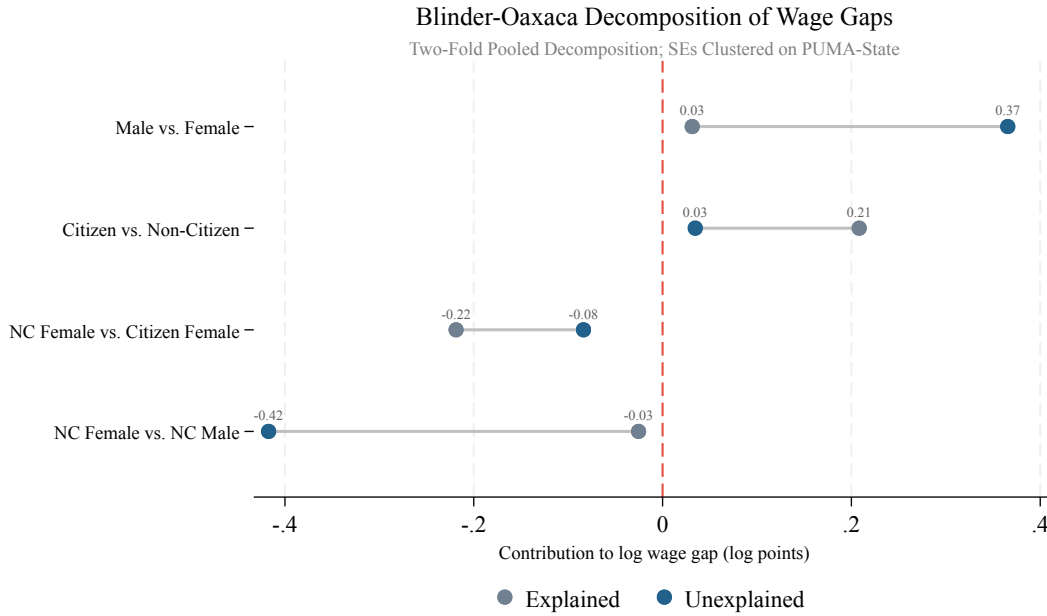
Standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

All decompositions are unweighted, in log points, and control for centered age and its square, education group, English fluency group, children group, and year and industry indicators. “NC” denotes non-citizen. The gap is the first-listed group’s mean log wage minus the second’s.

The decompositions reveal that the gender and citizenship gaps have opposite anatomies, a contrast that Figure 6 makes visible at a glance: in the two gender-based contrasts (the first and last rows) the navy unexplained dot sits far from zero while the gray explained dot hugs it, whereas the citizenship contrast displays the mirror image. The gender gap of 0.397 log points (women

Fig. 6: Blinder–Oaxaca Decompositions: Explained vs. Unexplained Components



*Note:* Dumbbell plot of the two-fold pooled decompositions reported in Table 2. Each horizontal segment connects a contrast’s explained component (gray dot) to its unexplained component (navy dot).

earn roughly 33% less) is almost entirely—92.1%—an unexplained wage penalty: differences in returns, not differences in qualifications. Indeed, the explained component is small (0.031) only because two forces offset: women’s industry composition explains 0.071 log points of the gap, while women’s *advantage* in educational attainment works in the opposite direction ( $-0.039$ ). Within the unexplained component, the largest identifiable contribution comes from returns associated with children (0.106 log points): the presence of children raises men’s wages more than women’s, a fatherhood premium coexisting with a motherhood penalty. This is the signature of statistical and taste-based discrimination rather than a human capital deficit.

The citizenship gap of 0.243 log points (about 21.6%) shows the reverse pattern: 85.7% is an explained wage differential, driven overwhelmingly by English proficiency (0.137 log points) and educational attainment (0.088). The remaining unexplained component (0.035,  $p < 0.05$ ) is economically small. Citizenship-based disparities thus operate primarily through structural gaps in human capital accumulation, consistent with Coulombe et al. (2014), rather than through unequal pay for equally qualified workers.

The two intersectional contrasts show how these forces compound for non-citizen women. Rel-

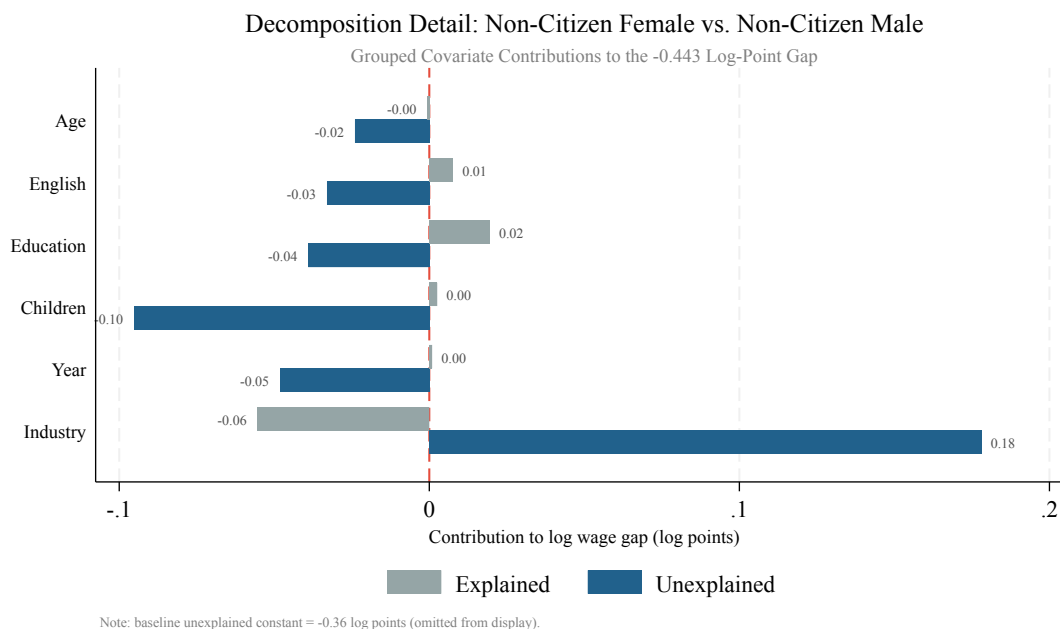


ative to citizen women, non-citizen women earn approximately 26.1% less; roughly 72% of this gap reflects their lower English proficiency and education, but a statistically significant 27.7% reflects lower returns to the characteristics they do hold, including significantly lower returns to English proficiency and education—direct evidence of the credential devaluation emphasized by Chiswick (1978) and Blau and Kahn (2015). Most strikingly, relative to non-citizen men, non-citizen women earn approximately 35.8% less, a larger gap than the economy-wide gender gap, even though their observable characteristics are, on net, nearly identical (explained share: just 5.7%, and their education advantage partially offsets their industry composition). Fully 94.3% of this gap is unexplained, with the returns associated with children again contributing heavily. Non-citizen women therefore face the labor market’s most severe differential treatment: penalized once through structural human capital barriers relative to citizen women, and again through lower returns to their observed characteristics relative to non-citizen men.

Figure 7 unpacks this flagship contrast by covariate block. The explained components are relatively small: non-citizen women’s education advantage (+0.019) partially offsets their less favorable industry composition (−0.055). In contrast, the unexplained component accounts for most of the gap, with negative contributions from children (−0.095), education (−0.039), English proficiency (−0.033), and age (−0.024), partially offset by industry returns (+0.178). The remaining unexplained constant (−0.356 log points), omitted from the figure for readability, captures the wage difference between non-citizen women and non-citizen men with identical observed characteristics.

To synthesize how the two methods speak to each other, Table 3 compares the regression and decomposition evidence. The Mincer regression measures conditional wage differentials, while the decomposition attributes the raw gaps to characteristics versus returns—the latter often interpreted as structural discrimination.

Fig. 7: Decomposition Detail: Non-Citizen Female vs. Non-Citizen Male



*Note:* Grouped covariate contributions to the approximately 36% wage gap decomposed in Table 11.. The unexplained baseline constant is omitted from the display so the smaller covariate contributions remain legible.

Table 3: Mincer Regression and Oaxaca Decomposition: Comparison

| Category                  | Mincer Regression                                                                                        | Oaxaca Decomposition                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Purpose</b>            | Estimates conditional association between predictors and log wages                                       | Decomposes raw wage gap into “explained” vs. “unexplained” parts                 |
| <b>Focus</b>              | Magnitude of wage differentials by group, holding controls fixed                                         | How much of each differential is due to characteristics vs. differential returns |
| <b>Gender Result</b>      | Women earn $\approx 30\%$ less than men in the preferred specification                                   | 92.1% of the raw gap is unexplained                                              |
| <b>Citizenship Result</b> | No significant non-citizen penalty remains after controls; a modest 4.6% naturalization premium persists | 85.7% of the raw gap is explained by human capital (English, education)          |
| <b>Intersection</b>       | NC women earn approximately 4.7% less beyond the separate gender and citizenship terms                   | 94.3% of the non-citizen gender gap is unexplained                               |
| <b>Limitations</b>        | Coefficients are conditional associations; unobservables remain                                          | Unexplained part includes omitted variables, not just discrimination             |

Returning to the hypotheses introduced in the theoretical section, the evidence rejects the null hypothesis that non-citizen women experience no wage gap once human capital differences are accounted for. Across all three regression specifications the non-citizen  $\times$  female interaction is negative and statistically significant, and the decompositions show that the gaps facing non-citizen women are the least attributable to observable characteristics of any contrast examined. The results therefore support the alternative hypothesis: non-citizen women face compounded penalties beyond the independent associations of gender and citizenship alone.

## C Robustness

Two features of the design bear on the credibility of these findings. First, because the sample is a single-stage cluster sample of geographies, all standard errors are clustered on the 118 PUMA–state units; the PUMA–state fixed effects are nested within these clusters, so the reported inference already accounts for arbitrary within-geography correlation. Second, the main estimates are unweighted; Appendix Tables 13 and 14 re-estimate the preferred regression and all four decompositions using ACS person weights (`perwt`). The results are substantively unchanged: the weighted gender penalty is 28.6% (30.1% unweighted), the weighted additional penalty for non-citizen women is 5.1% (4.7% unweighted), and both remain significant at the 1% level. The weighted decompositions reproduce the central asymmetry: the gender gap remains 93% unexplained, while the citizenship gap remains 87% explained. If anything, incorporating the weights strengthens the main conclusions.

Figure 8 provides a final specification diagnostic: residuals plotted against fitted values from the preferred model. The residual spread is roughly constant across the range of fitted log wages, with no fan-shaped pattern that would signal severe heteroskedasticity tied to the conditional mean; the longer lower tail simply reflects the left skew that remains in log wages (skewness  $-1.09$ ) from a small number of very low reported earnings. In any case, no inference in this paper relies on homoskedasticity: every reported standard error is cluster-robust at the PUMA–state level, which remains valid under arbitrary heteroskedasticity and within-geography error correlation.

Fig. 8: Model Diagnostic: Residuals vs. Fitted Log Wages



*Note:* Residuals and fitted values from Model 3. A 5% random subsample of the 526,832 observations is displayed for legibility.

## VII Conclusion and Discussion

Immigrants are economically indispensable yet persistently undercompensated by the U.S. labor market. The results highlight a troubling reality: citizenship status and gender intersect to exacerbate wage disparities, impacting non-citizen women the most. While human capital factors largely explain wage differences across citizenship groups, a conditional gender gap of roughly 30% persists in the preferred specification—and the decomposition attributes 92% of the raw gender gap to unequal returns rather than unequal qualifications, highlighting the structural devaluation of women’s labor. Echoing Oreopoulos (2011), these disparities appear driven largely by intangible factors such as employer bias and statistical discrimination, where gender acts as a proxy for unobserved productivity (Aigner and Cain, 1977).

The raw non-citizen wage penalty of 22% disappears once human capital and race controls and fixed effects are introduced, aligning closely with Coulombe et al. (2014), who attribute citizenship gaps primarily to differences in education, skills, and language rather than overt discrimination. However, the modest but significant naturalization premium of 4.6% reinforces Donahue (2021)

and Borjas (1994), indicating that legal status improves labor market outcomes by mitigating, though not eliminating, structural barriers. The premium suggests tangible economic benefits from naturalization, such as enhanced labor market access and greater legal protections, as highlighted by Blau and Kahn (2015). Nevertheless, naturalization alone does not close the disparities rooted in gender.

Intersectionality emerges as a crucial determinant of wage outcomes. Non-citizen women face an additional penalty of about 4.7% beyond what gender and citizenship status alone would predict, and within the non-citizen workforce women earn roughly 36% less than men—a gap that exceeds the economy-wide gender gap, with 94% of it unexplained by observable characteristics. This compounded disadvantage illustrates the “double jeopardy” described by Browne and Misra (2003) and aligns with Vernez’s (1999) findings on heightened cultural and institutional barriers uniquely faced by immigrant women. The racial gradients estimated alongside these effects—most notably an 18% conditional penalty for Black workers relative to observationally similar White workers—underscore that racial stratification operates on top of, not instead of, citizenship and gender.

Future research should explore credential devaluation by disaggregating foreign education by country or region, distinguishing, for example, the wage returns to Mexican versus Southeast Asian degrees. Additionally, separating non-citizens by visa category (e.g., H-1B versus H-2A) could uncover heterogeneity in labor market outcomes obscured in aggregate analyses. Finally, targeted policy proposals such as incentives for firms to adopt anti-bias training in high-immigrant sectors deserve empirical evaluation, especially given the feasibility constraints imposed by current U.S. labor law.

The devaluation of female labor exposes the entrenched sexism and xenophobia that persist in the U.S. labor market. When these forces intersect, they compound wage penalties for non-citizen immigrant women. To combat these disparities, policymakers must address discrimination explicitly. Structural reforms must go beyond improving access to human capital; they must dismantle the biases embedded in labor market institutions. Only then can immigrant labor cease to be viewed as an *undesired necessity* and become fully recognized as what it is: a fundamental driver of U.S. economic growth deserving of equal opportunity and equal reward.

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## Appendix

### A Descriptive Statistics

Table 4: Distribution of Citizenship Status in the Sample

| <b>Citizenship Status</b> | <b>Freq.</b> | <b>Percent</b> | <b>Cum.</b> |
|---------------------------|--------------|----------------|-------------|
| Native Citizen            | 448,681      | 85.17          | 85.17       |
| Naturalized Citizen       | 43,450       | 8.25           | 93.41       |
| Non-Citizen               | 34,701       | 6.59           | 100.00      |
| <b>Total</b>              | 526,832      | 100.00         |             |

Table 5: Annual Mean Wages by Sex and Citizenship (\$)

| <b>Citizenship Status</b> | <b>Male</b> | <b>Female</b> | <b>Total</b> |
|---------------------------|-------------|---------------|--------------|
| Native Citizen            | 63,130.29   | 41,446.54     | 52,516.93    |
| Naturalized Citizen       | 69,754.73   | 47,531.38     | 58,556.11    |
| Non-Citizen               | 48,421.68   | 31,863.99     | 41,713.38    |
| <b>Total</b>              | 62,537.43   | 41,440.69     | 52,303.41    |

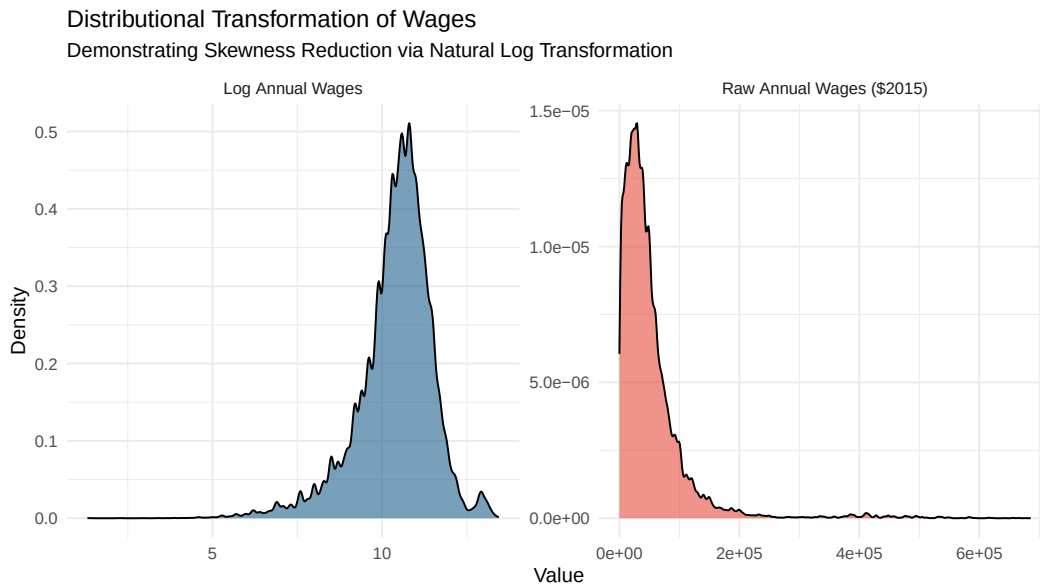
Table 6: Annual Median Wages by Sex and Citizenship (\$)

| <b>Citizenship Status</b> | <b>Male</b> | <b>Female</b> | <b>Total</b> |
|---------------------------|-------------|---------------|--------------|
| Native Citizen            | 46,361.31   | 31,590.33     | 38,634.43    |
| Naturalized Citizen       | 48,068.28   | 34,551.92     | 40,056.90    |
| Non-Citizen               | 30,000.00   | 20,748.22     | 26,000.00    |
| <b>Total</b>              | 45,000.00   | 30,981.51     | 37,724.04    |

Table 7: Industry Distribution for Non-Citizens

| Industry Name          | Freq.         | Percent       | Cum.   |
|------------------------|---------------|---------------|--------|
| Agriculture & Forestry | 885           | 2.55          | 2.55   |
| Arts                   | 4,601         | 13.26         | 15.81  |
| Construction           | 3,498         | 10.08         | 25.89  |
| Education & Health     | 5,669         | 16.34         | 42.23  |
| Finance & Real Estate  | 1,516         | 4.37          | 46.60  |
| Information            | 477           | 1.37          | 47.97  |
| Manufacturing          | 4,867         | 14.03         | 62.00  |
| Military               | 8             | 0.02          | 62.02  |
| Mining & Oil           | 204           | 0.59          | 62.61  |
| Other Services         | 1,913         | 5.51          | 68.12  |
| Professional Services  | 4,779         | 13.77         | 81.89  |
| Public Administration  | 395           | 1.14          | 83.03  |
| Retail                 | 3,234         | 9.32          | 92.35  |
| Transportation         | 1,600         | 4.61          | 96.96  |
| Wholesale              | 1,055         | 3.04          | 100.00 |
| <b>Total</b>           | <b>34,701</b> | <b>100.00</b> |        |

Fig. 9: Distributional Transformation of Wages

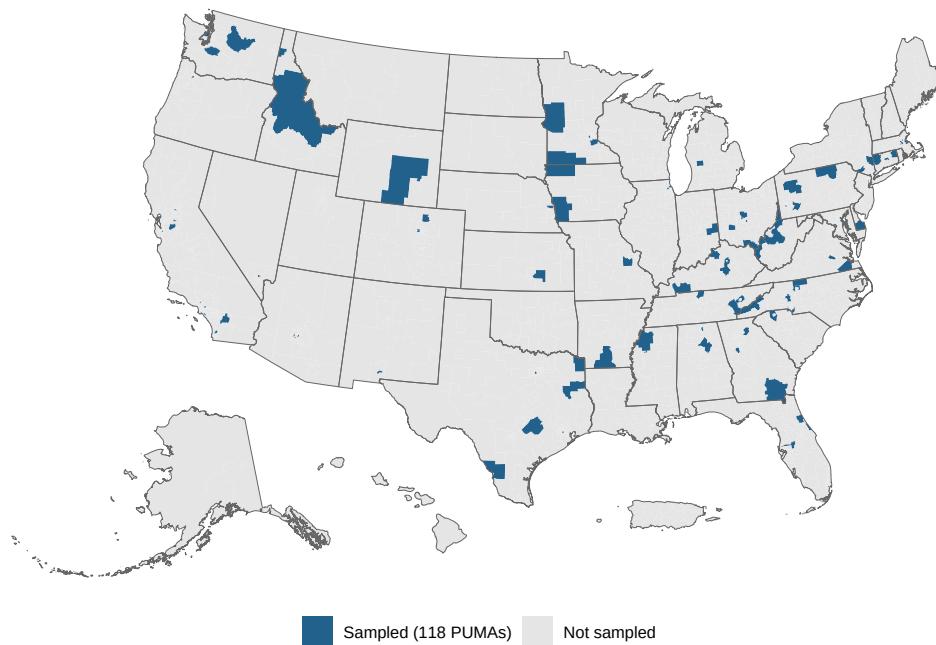


*Note:* Kernel density of annual wages in 2015 dollars (right) and of log wages (left). The log transformation reduces skewness from 4.22 to  $-1.09$ , motivating the log wage dependent variable in Equation (2).

Fig. 10: Geographic Coverage of the 5% PUMA–State Cluster Sample

**Geographic Coverage of the 5% PUMA–State Cluster Sample**

2010–Definition PUMAs; Sampled Geographies Shaded (ACS 2012–2019)



*Note:* 2010-definition PUMA boundaries from U.S. Census Bureau TIGER files (accessed via the **tigris** R package). Shaded areas are the 118 PUMA–state geographies drawn by the 5% single-stage cluster sample; Alaska, Hawaii, and Puerto Rico are rescaled for display. Sampled geographies span all four census regions and include both metropolitan and rural areas.

## B Detailed Blinder–Oaxaca Analysis

Each decomposition below reports the two-fold pooled decomposition of Equation (3): group mean log wages, the total gap, and the explained and unexplained components, followed by the contribution of each covariate block (age, English fluency, education, children, year, and industry indicators) to both components. The unexplained constant absorbs the baseline difference in intercepts between the groups relative to the pooled model. All decompositions are unweighted with standard errors clustered on the 118 PUMA–state geographies. Components are reported in log points, which are additive across the decomposition; the % column converts each overall component to its percentage-differential equivalent,  $100(e^\beta - 1)$ .

### 1 Gender Wage Decomposition

Table 8: Gender Oaxaca Decomposition (Male vs. Female)

|             | Overall   |         |       | Contributions |           |             |
|-------------|-----------|---------|-------|---------------|-----------|-------------|
|             | Coef.     | SE      | %     | Block         | Explained | Unexplained |
| Male        | 10.576*** | (0.020) | —     | Age           | 0.004**   | 0.000       |
| Female      | 10.179*** | (0.018) | —     | English       | −0.003*** | −0.005***   |
| Wage Gap    | 0.397***  | (0.010) | +48.7 | Education     | −0.039*** | 0.008       |
| Explained   | 0.031***  | (0.005) | +3.2  | Children      | −0.002*** | 0.106***    |
| Unexplained | 0.366***  | (0.008) | +44.1 | Year          | 0.000**   | 0.001       |
|             |           |         |       | Industry      | 0.071***  | −0.187***   |
|             |           |         |       | Constant      | —         | 0.442***    |

Clustered standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 8 decomposes the 0.397 log-point gender gap (men earn 48.7% more than women, or equivalently women earn 32.8% less). The explained component (0.031, a 3.2% differential) accounts for only 7.9% of the gap, and its composition is instructive: industry sorting explains 0.071 log points, but women’s *higher* educational attainment offsets this with a −0.039 contribution—women are, on observables, at least as qualified as men. The unexplained component (0.366) accounts for 92.1% of the gap. Its largest identifiable piece is the differential return associated with children (0.106): children are associated with higher wages for men and lower wages for women, the classic fatherhood premium/motherhood penalty asymmetry. The positive constant (0.442) captures the remaining baseline difference in intercepts—pay differences between men and women with identical

measured characteristics—consistent with statistical discrimination, whereby employers systematically undervalue women’s productivity despite equivalent qualifications.

## 2 Citizenship Status Decomposition

Table 9: Citizenship Oaxaca Decomposition (Citizen vs. Non-Citizen)

|              | Overall   |         |       | Contributions |           |             |
|--------------|-----------|---------|-------|---------------|-----------|-------------|
|              | Coef.     | SE      | %     | Block         | Explained | Unexplained |
| Citizens     | 10.399*** | (0.018) | —     | Age           | −0.020*** | −0.047***   |
| Non-Citizens | 10.156*** | (0.031) | —     | English       | 0.137***  | 0.035***    |
| Wage Gap     | 0.243***  | (0.023) | +27.5 | Education     | 0.088***  | 0.034***    |
| Explained    | 0.208***  | (0.021) | +23.2 | Children      | −0.004**  | 0.023***    |
| Unexplained  | 0.035**   | (0.014) | +3.5  | Year          | 0.001***  | 0.003       |
|              |           |         |       | Industry      | 0.006     | 0.154***    |
|              |           |         |       | Constant      | —         | −0.168***   |

Clustered standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 9 decomposes the 0.243 log-point gap between citizens and non-citizens (citizens earn 27.5% more, or non-citizens 21.6% less). In sharp contrast to the gender decomposition, the explained component (0.208, a 23.2% differential) accounts for 85.7% of the gap. English proficiency is the single largest explained contributor (0.137), followed by educational attainment (0.088): non-citizens’ wage disadvantage operates primarily through lower stocks of these forms of human capital, an explained wage differential rather than an unexplained penalty. The unexplained component (0.035) is economically small though statistically significant at the 5% level. This anatomy supports the human capital reading of the citizenship gap—structural inequality in access to education and language acquisition—rather than differential treatment of equally qualified workers.

## 3 Non-Citizen Females vs. Citizen Females

Table 10 examines the citizenship dimension within the female workforce. Non-citizen women earn 0.302 log points (about 26%) less than citizen women. The explained component (−0.219, 72.3% of the gap) is dominated by non-citizen women’s lower English proficiency (−0.104) and educational attainment (−0.090), with adverse industry composition (−0.037) adding to the burden. But the unexplained component (−0.084, 27.7%) is also statistically significant, and its composition matters: the returns to English proficiency (−0.077) and to education (−0.059) are significantly

Table 10: Non-Citizen Females vs. Citizen Females Oaxaca Decomposition

|                    | Overall   |         |       | Contributions |           |             |
|--------------------|-----------|---------|-------|---------------|-----------|-------------|
|                    | Coef.     | SE      | %     | Block         | Explained | Unexplained |
| Non-Citizen Female | 9.893***  | (0.031) | —     | Age           | 0.031***  | 0.027**     |
| Citizen Female     | 10.195*** | (0.018) | —     | English       | −0.104*** | −0.077***   |
| Wage Gap           | −0.302*** | (0.024) | −26.1 | Education     | −0.090*** | −0.059***   |
| Explained          | −0.219*** | (0.022) | −19.6 | Children      | −0.017*** | 0.006       |
| Unexplained        | −0.084*** | (0.015) | −8.0  | Year          | −0.001    | −0.026      |
|                    |           |         |       | Industry      | −0.037*** | −0.153      |
|                    |           |         |       | Constant      | —         | 0.199*      |

Clustered standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ 

lower for non-citizen women than the pooled benchmark. This is the empirical footprint of credential devaluation: non-citizen women are penalized both for possessing fewer credentials and for receiving less value from the credentials they hold.

#### 4 Non-Citizen Females vs. Non-Citizen Males

Table 11: Non-Citizen Females vs. Non-Citizen Males Oaxaca Decomposition

|                    | Overall   |         |       | Contributions |           |             |
|--------------------|-----------|---------|-------|---------------|-----------|-------------|
|                    | Coef.     | SE      | %     | Block         | Explained | Unexplained |
| Non-Citizen Female | 9.893***  | (0.031) | —     | Age           | −0.001    | −0.024**    |
| Non-Citizen Male   | 10.336*** | (0.034) | —     | English       | 0.008**   | −0.033***   |
| Wage Gap           | −0.443*** | (0.017) | −35.8 | Education     | 0.019***  | −0.039***   |
| Explained          | −0.025**  | (0.010) | −2.5  | Children      | 0.003**   | −0.095***   |
| Unexplained        | −0.417*** | (0.015) | −34.1 | Year          | 0.001*    | −0.048      |
|                    |           |         |       | Industry      | −0.055*** | 0.178*      |
|                    |           |         |       | Constant      | —         | −0.356***   |

Clustered standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ 

Table 11 isolates the gender dimension within the non-citizen population and delivers the paper's most striking result. The gap of  $-0.443$  log points (about 36%) exceeds the economy-wide gender gap, yet the explained component is tiny ( $-0.025$ , 5.7%): non-citizen women's education advantage over non-citizen men (0.019) nearly offsets their less favorable industry composition ( $-0.055$ ). Fully 94.3% of the gap is unexplained. The returns associated with children ( $-0.095$ ) again penalize women, and the large negative constant ( $-0.356$ ) indicates that non-citizen women earn substantially less than non-citizen men with identical measured characteristics. Despite equiv-

alent or superior qualifications, non-citizen women receive systematically lower returns—the most severe differential treatment observed across all four contrasts.

## 5 Synthesis of Decomposition Findings

Three regularities emerge. First, gender-based wage gaps are predominantly unexplained wage penalties: the labor market assigns lower returns to women’s qualifications than to men’s, a pattern that holds economy-wide (92.1% unexplained) and intensifies within the non-citizen workforce (94.3%). Second, citizenship-based wage gaps are predominantly explained wage differentials, rooted in observable human capital disparities—especially English proficiency and education—consistent with structural disadvantage in human capital accumulation rather than unequal treatment of equally qualified workers. Third, intersectionality amplifies disadvantage multiplicatively: non-citizen women are penalized through both channels at once, facing human capital barriers relative to citizen women and discriminatory returns relative to non-citizen men. These findings align with statistical discrimination theory, in which employers use gender and citizenship as productivity heuristics, penalizing marginalized workers even when qualifications are equal or superior.

## C Robustness Checks



Table 12: Full Coefficient Estimates, All Models

| Variables                        | Model 1              | Model 2              | Model 3              | % (M3) |
|----------------------------------|----------------------|----------------------|----------------------|--------|
| Naturalized Citizen (NC)         | 0.148***<br>(0.028)  | 0.042***<br>(0.013)  | 0.045**<br>(0.019)   | +4.6   |
| Non-Citizen (NotC)               | -0.247***<br>(0.030) | -0.018<br>(0.017)    | -0.020<br>(0.020)    | -2.0   |
| Female                           | -0.403***<br>(0.012) | -0.437***<br>(0.012) | -0.359***<br>(0.009) | -30.1  |
| NC $\times$ Female               | 0.019<br>(0.021)     | 0.043**<br>(0.021)   | 0.023<br>(0.018)     | +2.4   |
| NotC $\times$ Female             | -0.040**<br>(0.018)  | -0.042***<br>(0.016) | -0.048***<br>(0.014) | -4.7   |
| Age (centered)                   |                      | 0.022***<br>(0.001)  | 0.020***<br>(0.000)  | —      |
| Age <sup>2</sup> (centered)      |                      | -0.002***<br>(0.000) | -0.002***<br>(0.000) | —      |
| Education: Some College          |                      | 0.242***<br>(0.005)  | 0.211***<br>(0.005)  | +23.5  |
| Education: Bachelor's or Higher  |                      | 0.744***<br>(0.024)  | 0.621***<br>(0.010)  | +86.1  |
| English: Intermediate            |                      | -0.231***<br>(0.014) | -0.189***<br>(0.010) | -17.2  |
| English: Limited                 |                      | -0.396***<br>(0.021) | -0.337***<br>(0.018) | -28.6  |
| Children: One                    |                      | 0.069***<br>(0.009)  | 0.070***<br>(0.004)  | +7.3   |
| Children: Two                    |                      | 0.070***<br>(0.014)  | 0.073***<br>(0.005)  | +7.6   |
| Children: Three-Four             |                      | 0.017<br>(0.015)     | 0.033***<br>(0.007)  | +3.4   |
| Children: Five or more           |                      | -0.068***<br>(0.024) | -0.035*<br>(0.019)   | -3.5   |
| Race: Black                      |                      | -0.182***<br>(0.022) | -0.200***<br>(0.010) | -18.2  |
| Race: Amer. Indian/Alaska Native |                      | -0.159***<br>(0.027) | -0.100***<br>(0.033) | -9.5   |
| Race: Asian/Pacific Islander     |                      | 0.018<br>(0.020)     | -0.050***<br>(0.016) | -4.9   |
| Race: Hispanic/Latino            |                      | -0.035**<br>(0.015)  | -0.069***<br>(0.009) | -6.7   |
| Constant                         | 10.583***<br>(0.024) | 10.591***<br>(0.015) | 10.577***<br>(0.007) | —      |
| Fixed Effects                    |                      |                      | X                    |        |
| Observations                     | 526,832              | 526,832              | 526,832              |        |
| $R^2$                            | 0.038                | 0.339                | 0.390                |        |

Standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ 

The % column converts the Model 3 indicator coefficients to percentage differentials.

Reference categories: citizen, male, high school or less, fluent English, no children, White.

Table 13: Model 3: Unweighted vs. Population-Weighted (ACS Person Weights)

| Variables                | Unweighted           | Weighted             |
|--------------------------|----------------------|----------------------|
| Naturalized Citizen (NC) | 0.045**<br>(0.019)   | 0.037*<br>(0.021)    |
| Non-Citizen (NotC)       | -0.020<br>(0.020)    | -0.011<br>(0.022)    |
| Female                   | -0.359***<br>(0.009) | -0.336***<br>(0.009) |
| NC $\times$ Female       | 0.023<br>(0.018)     | 0.014<br>(0.018)     |
| NotC $\times$ Female     | -0.048***<br>(0.014) | -0.053***<br>(0.015) |
| Constant                 | 10.577***<br>(0.007) | 10.560***<br>(0.007) |
| Observations             | 526,832              | 526,832              |
| $R^2$                    | 0.390                | 0.388                |

SE in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$   
 Both columns use the preferred controls and fixed effects of Table 1,  
 Model 3. The weighted column applies ACS person weights (`perwt`).

Table 14: Blinder–Oaxaca Decompositions, Population-Weighted

| Contrast                     | Total Gap            | Explained            | Unexplained          |
|------------------------------|----------------------|----------------------|----------------------|
| Male vs. Female              | 0.369***<br>(0.009)  | 0.026***<br>(0.005)  | 0.343***<br>(0.008)  |
| Citizen vs. Non-Citizen      | 0.244***<br>(0.019)  | 0.213***<br>(0.018)  | 0.031**<br>(0.014)   |
| NC Female vs. Citizen Female | -0.314***<br>(0.022) | -0.236***<br>(0.020) | -0.077***<br>(0.015) |
| NC Female vs. NC Male        | -0.427***<br>(0.018) | -0.022*<br>(0.011)   | -0.406***<br>(0.016) |

Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$   
 All decompositions weighted by ACS person weights (`perwt`); two-fold pooled decompositions  
 with the same controls as Table 2. Log points throughout; the total gaps correspond to  
 percentage differentials ( $100(e^\beta - 1)$ ) of +44.6%, +27.6%, -26.9%, and -34.8%, respectively.